

CS & IT ENGINEERING



Operating System

CPU Scheduling

Lecture -2

By- Vishvadeep Gothi sir



Recap of Previous Lecture



Topic

Process Scheduling

Topic

Process Scheduling Algorithms

Topic

FCFS Scheduling

Topics to be Covered



Topic

SJF Scheduling

Topic

SRTF Scheduling

Topic

LJF Scheduling



Topic : FCFS (First Come First Serve)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	24			
P2	0	3			
P3	0	3			



Topic : Convoy Effect



↓
FCFS suffers from Convoy effect





Topic : FCFS (First Come First Serve)

Advantages:

Easy to implement

No complex logic

No starvation

→ if a process waits for indefinite time
assumption $\Rightarrow$ Any process does not have infinite BT

Disadvantages:

No option of Preemption

Convoy effect makes the system slow



Topic : SJF (Shortest Job First)

Scheduling Criteria: *smallest BT process first*

Tie breaker
↓
FCFS

Type of Algorithm: *Non-preemptive*

check BT $\xrightarrow[\text{if equal}]{} \text{check AT}$ $\xrightarrow[\text{if equal}]{} \text{check id}$



Topic : SJF (Shortest Job First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	24	30	30	6
P2	0	3	3	3	0
P3	0	3	6	6	3



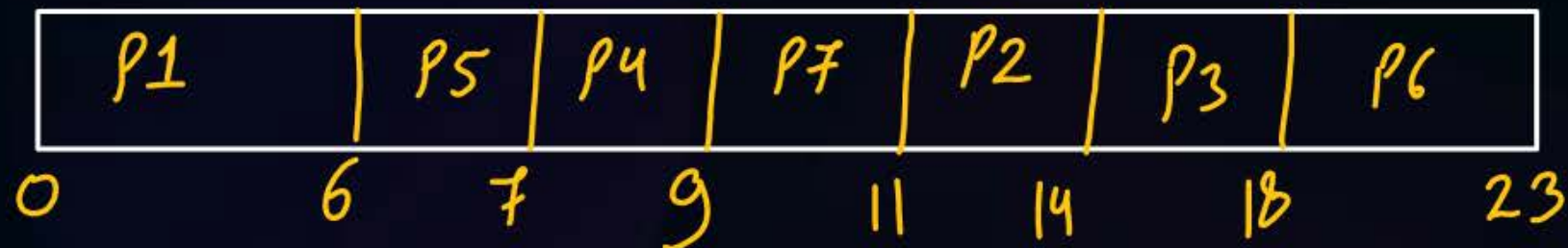
$$\text{avg TAT} = \frac{30 + 3 + 6}{3} = 13$$

$$\text{avg WT} = \frac{6 + 0 + 3}{3} = 3$$



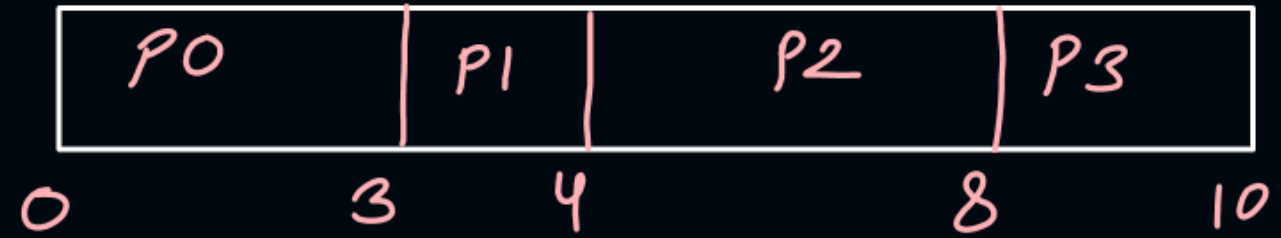
Topic : SJF (Shortest Job First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	6	6	6	0
P2	1	3	14	13	10
P3	2	4	18	16	12
P4	4	2	9	5	3
P5	5	1	7	2	1
P6	6	5	23	17	12
P7	8	2	11	3	1



$$\text{avg WT} = \frac{39}{7} = 5.57$$

Pid	AT	BT
P0	0	3
P1	1	1
P2	3	4
P3	6	2





Topic : SJF (Shortest Job First)

Advantages:

and avg TAT

1. Minimum average waiting time[^] among non-preemptive scheduling
2. Better throughput in continuous execution

Disadvantages:

1. No practical implementation because Burst time is not known in advance
2. No option of Preemption
3. Longer Processes may suffer from starvation

No. of context switches
in non-preemptive algo
for n processes $= n - 1$



Topic : SRTF (Shortest Remaining Time First)

↓
Preemptive version of STF

Scheduling Criteria: *smallest BT process first* | Tie breaker
↓
FCFS

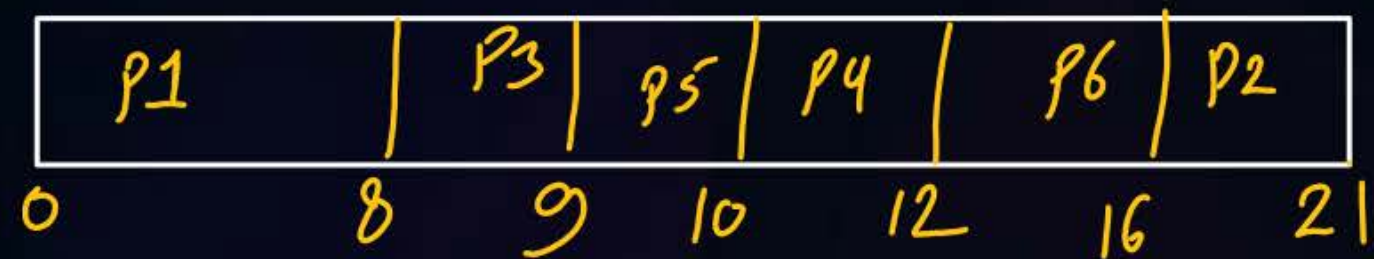
Type of Algorithm: *Preemptive*



Topic : SRTF (Shortest Job First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	8			
P2	1	5			
P3	2	1			
P4	3	2			
P5	4	1			
P6	5	4			

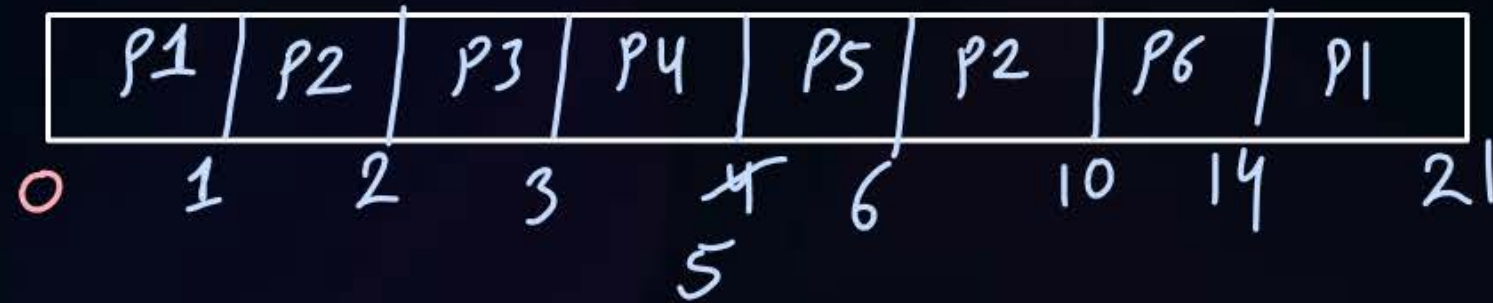
SJF





Topic : SRTF (Shortest Remaining Time First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	8	21		
P2	1	5	10		
P3	2	1	3		
P4	3	2	5		
P5	4	1	6		
P6	5	4	14		



No. of context switches = 7
(Ignore at 0 and 21)

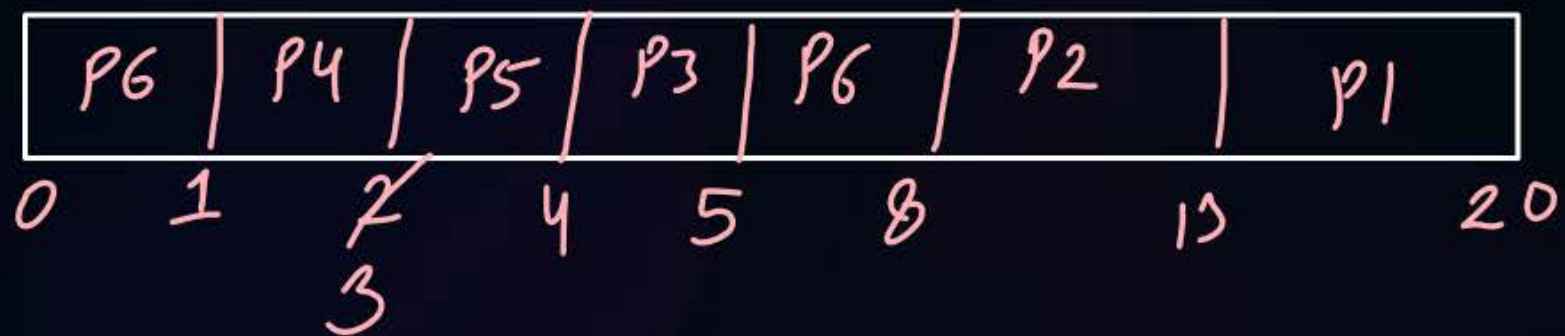
Time	Ready Queue
0	P1(8)
1	P1(7), P2(5)
2	P1(7), P2(4), P3(1)
3	P1(7), P2(4), P4(2)
4	P1(7), P2(4), P4(1), P5(1)
5	P1(7), P2(4), P5(1) , P6(4)

	BT
P1	8 7
P2	5 4
P3	1
P4	2 1
P5	1
P6	4



Topic : SRTF (Shortest Remaining Time First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	4	7			
P2	5	5			
P3	3	1			
P4	1	2			
P5	2	1			
P6	0	4			



no. of context switches = 6



Topic : SRTF (Shortest Remaining Time First)

H.W.

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	9			
P2	1	6			
P3	2	4			
P4	3	2			
P5	6	1			



Topic : SRTF (Shortest Remaining Time First)

Advantages:

& min. avg. TAT

1. Minimum average waiting time[^] among all scheduling algorithm
2. Better throughput in continue run

Disadvantages:

1. No practical implementation because Burst time is not known in advance
2. Longer Processes may suffer from starvation



2 mins Summary

Topic

SJF Scheduling

Topic

SRTF Scheduling

Topic

LF Scheduling



Happy Learning

THANK - YOU